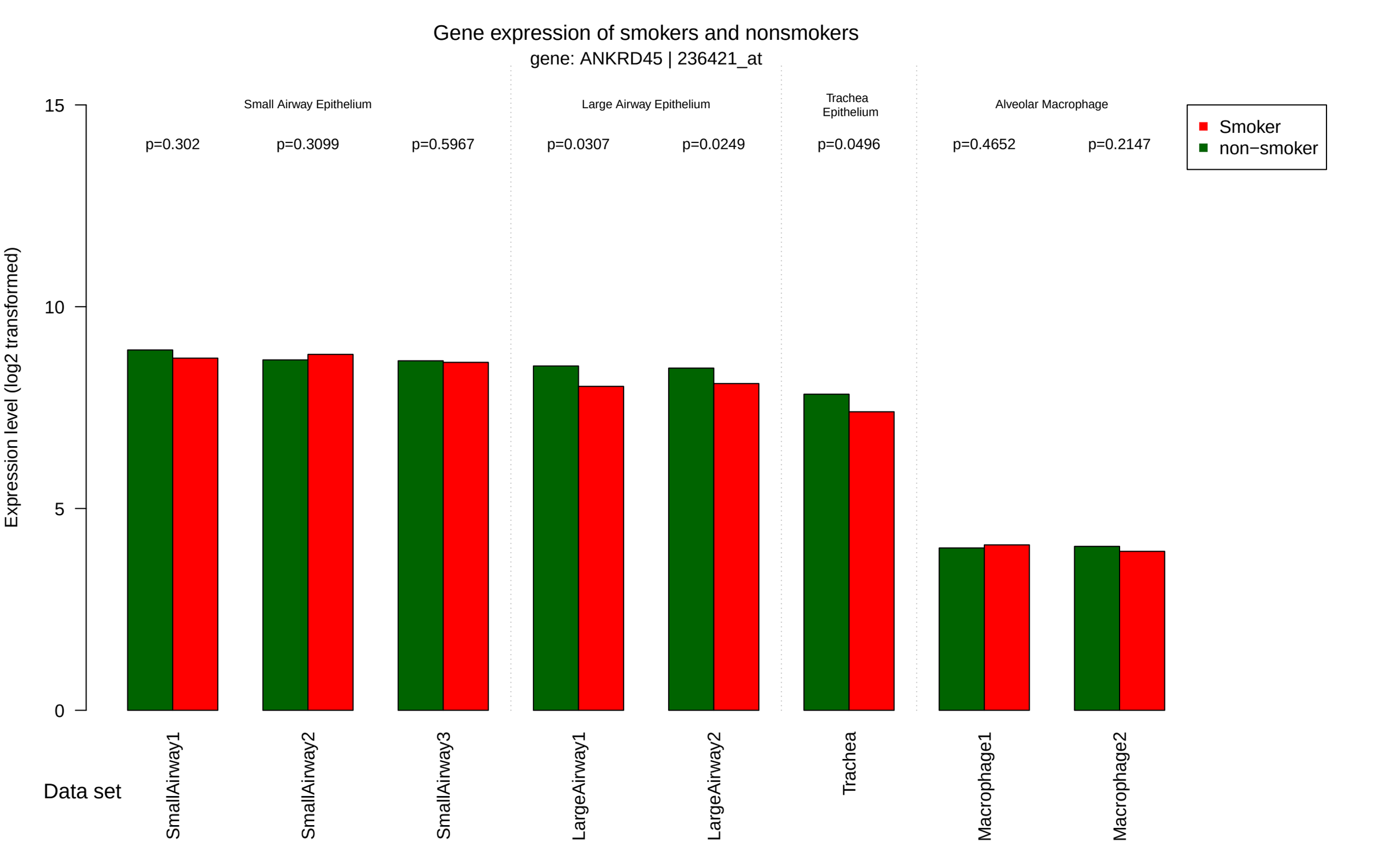
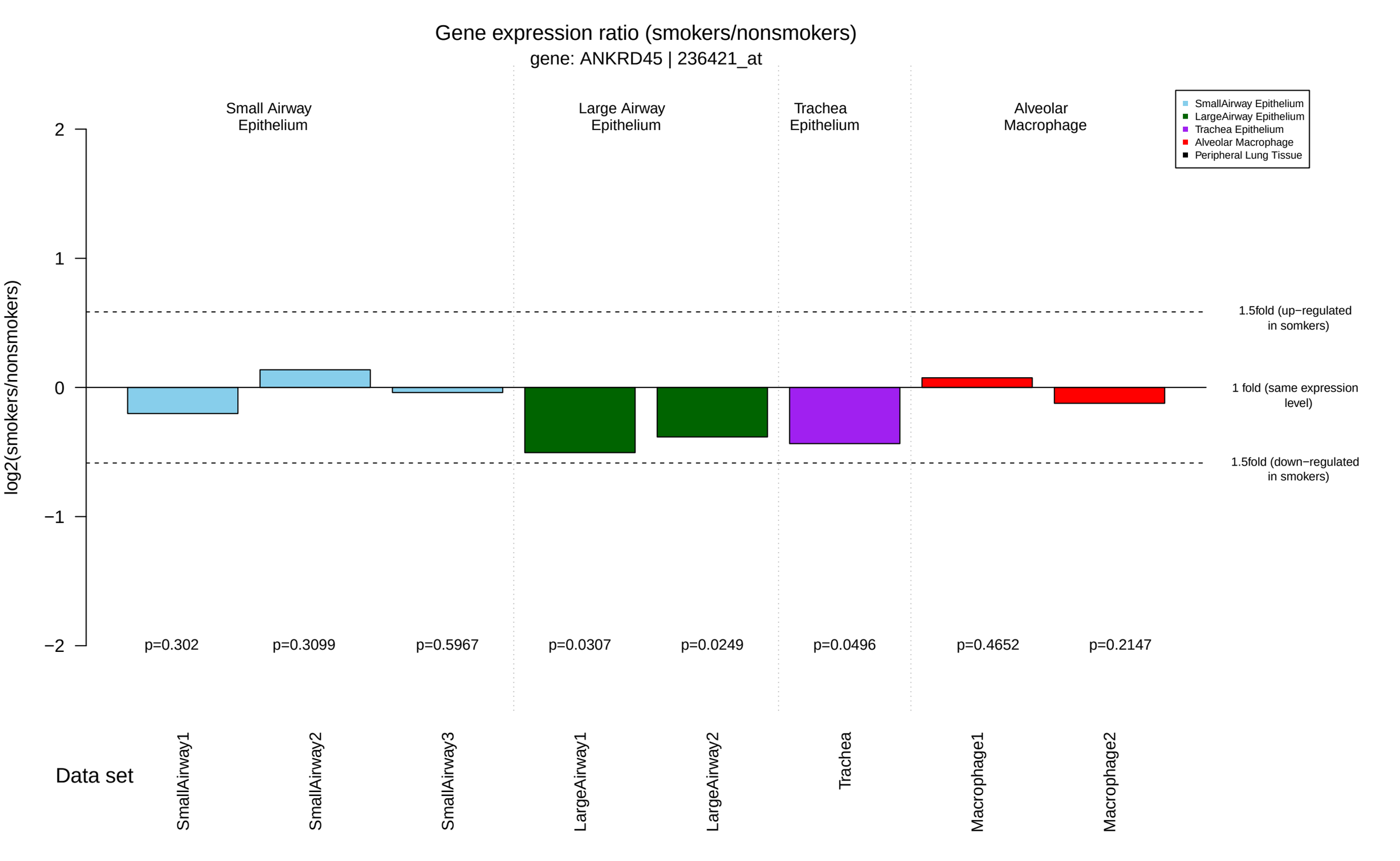
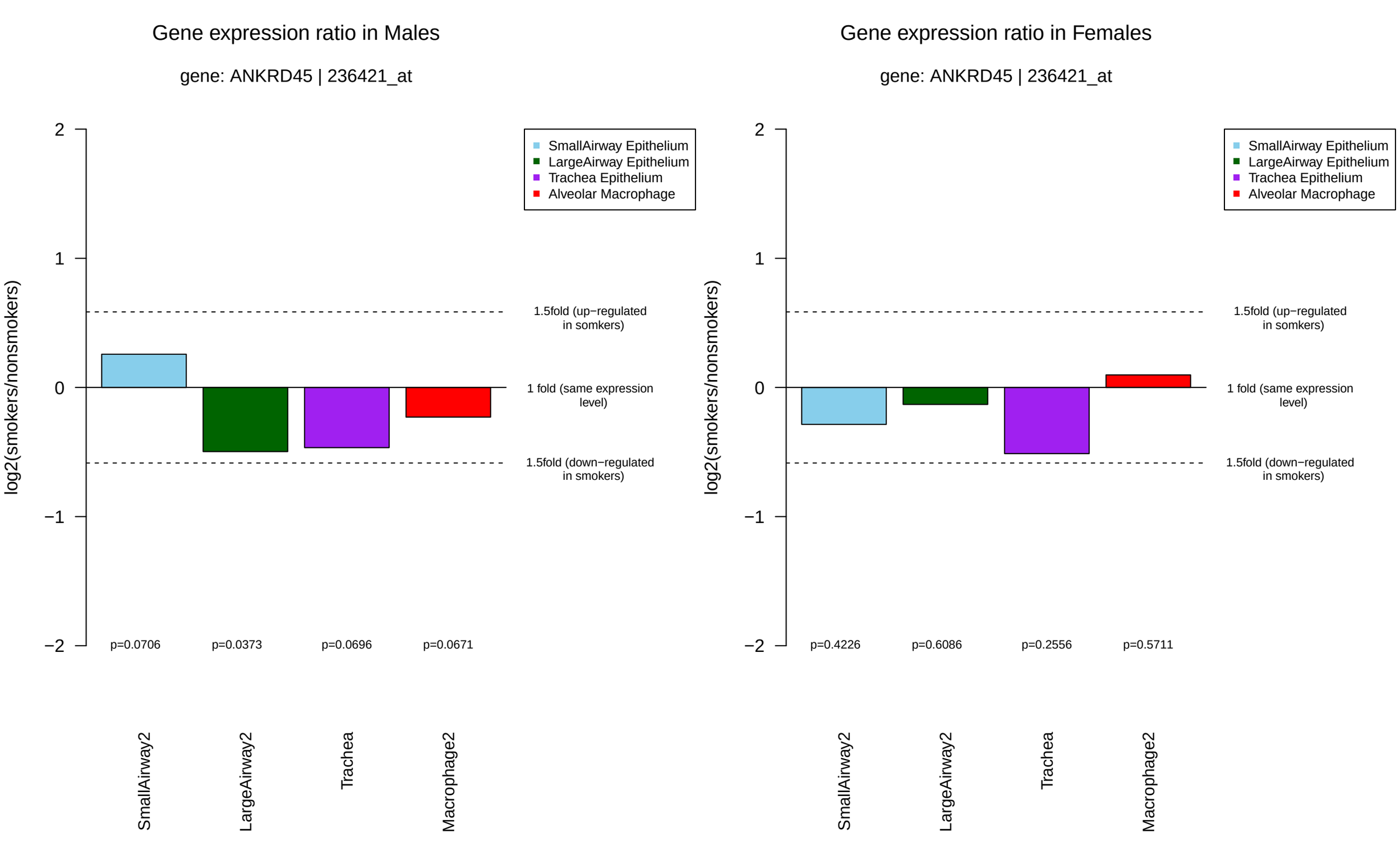
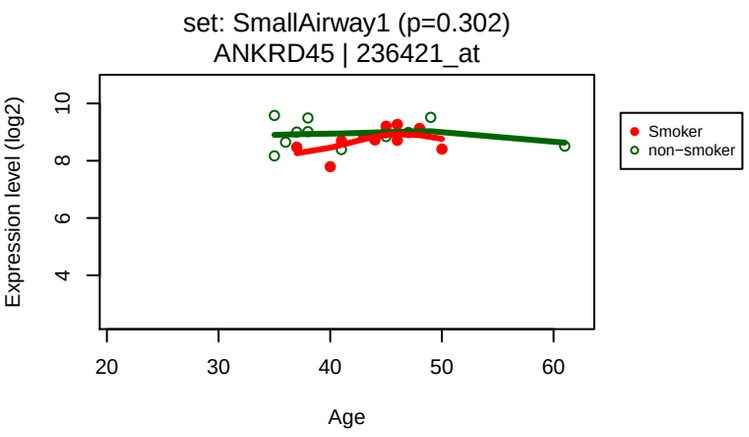
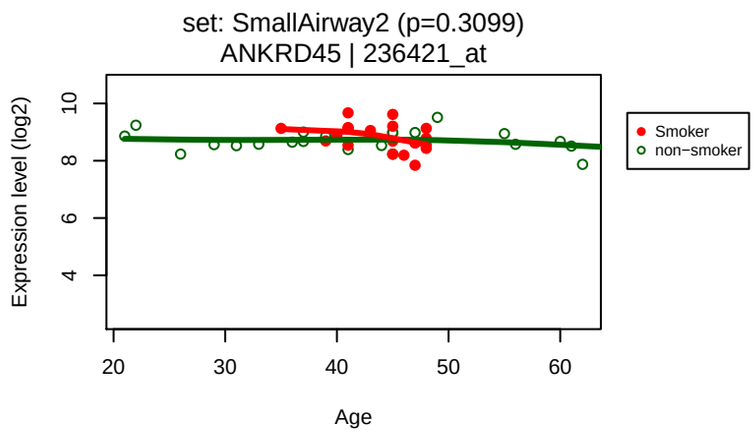
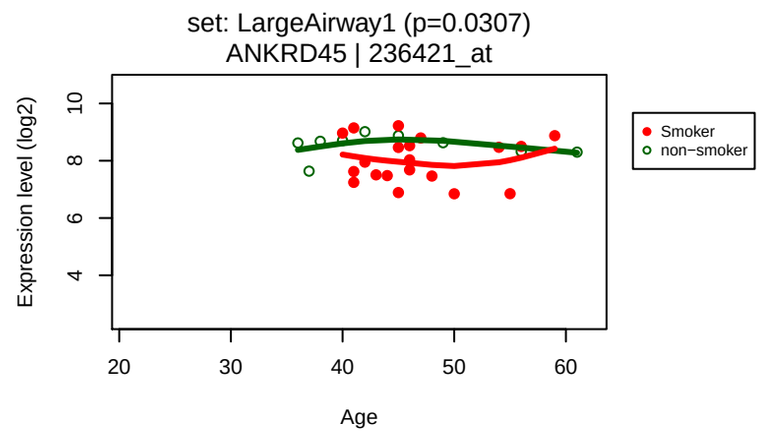

Gene expression of smokers and nonsmokers
gene: ANKRD45 | 236421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.302
p=0.3099
p=0.5967
p=0.0307
p=0.0249
p=0.0496
p=0.4652
p=0.2147
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

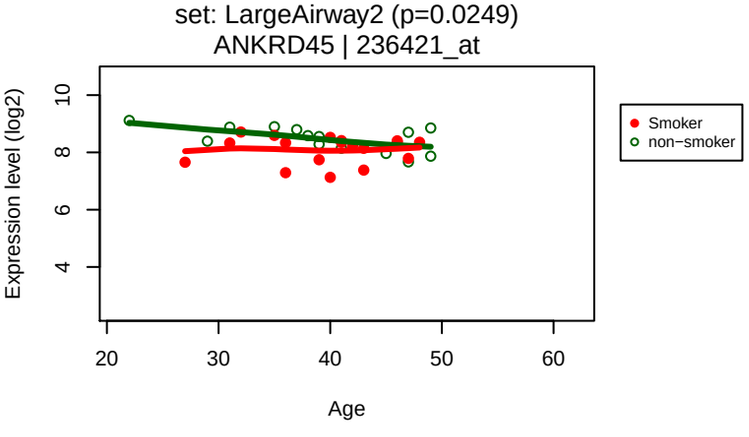

Gene expression ratio (smokers/nonsmokers)
gene: ANKRD45 | 236421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.302
p=0.3099
p=0.5967
p=0.0307
p=0.0249
p=0.0496
p=0.4652
p=0.2147
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

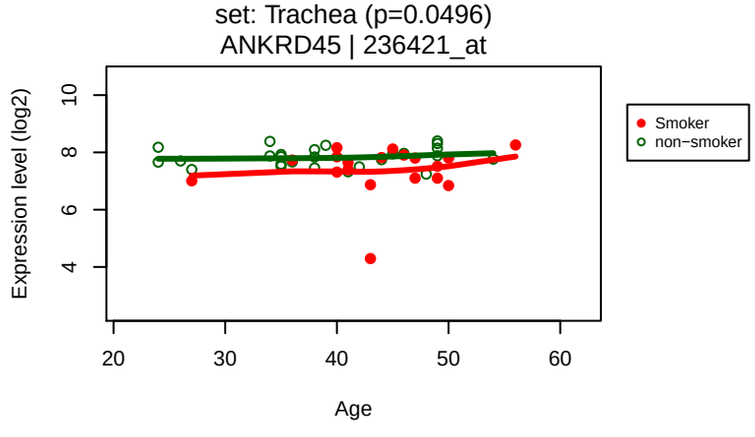

Gene expression ratio in Males
gene: ANKRD45 | 236421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0706
p=0.0373
p=0.0696
p=0.0671
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANKRD45 | 236421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4226
p=0.6086
p=0.2556
p=0.5711
SmallAirway2
LargeAirway2
Trachea
Macrophage2

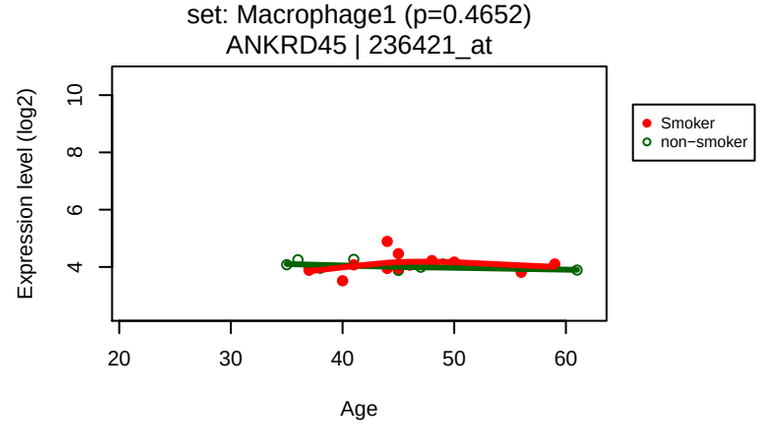

set: SmallAirway1 (p=0.302)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker

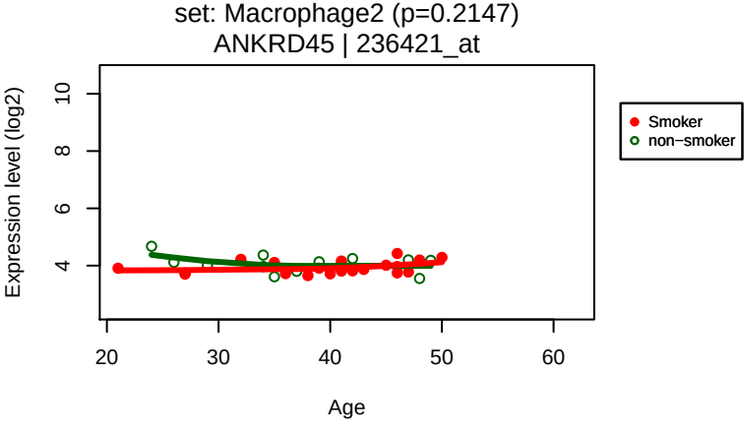

set: SmallAirway2 (p=0.3099)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker

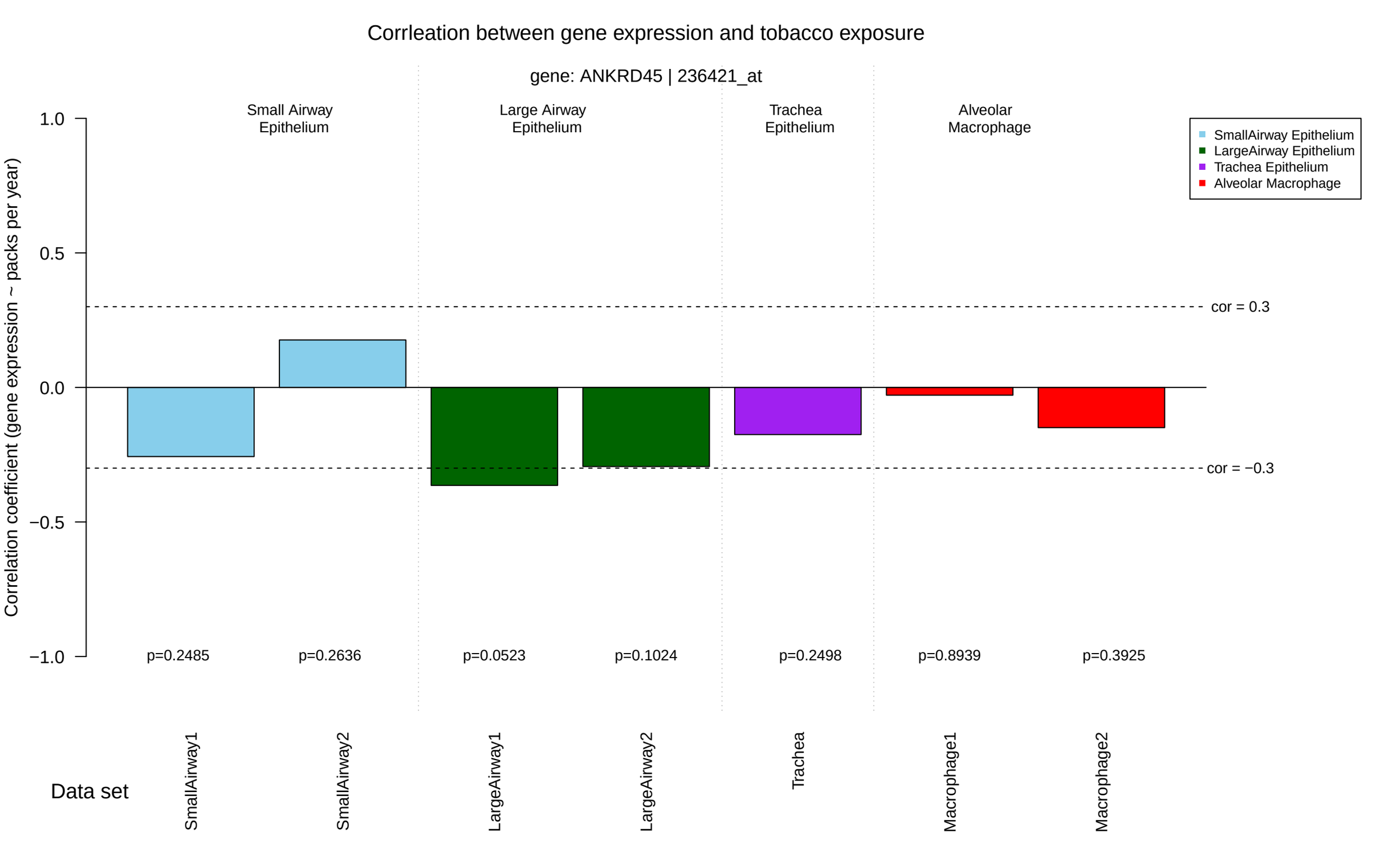

set: LargeAirway1 (p=0.0307)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0249)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0496)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4652)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2147)
ANKRD45 | 236421_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANKRD45 | 236421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2485
p=0.2636
p=0.0523
p=0.1024
p=0.2498
p=0.8939
p=0.3925
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2